

Module 03 - Groups & Memberships

Create and manage collaborative groups with roles

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Groups are the top-level collaboration container. Implement group CRUD, membership records, roles, member list, leave group, and owner/admin management.

Routes

```
/groups
/groups/:groupId
```

Group document

```
groups/{groupId}
{
  name,
  description,
  coverImageUrl,
  createdBy,
  createdAt,
  updatedAt,
  status
}
```

Membership document

```
groups/{groupId}/members/{uid}
{
  uid,
  role: "owner" | "admin" | "member",
  joinedAt,
  displayNameSnapshot,
  photoURLSnapshot
}
```

Permissions

Action	Owner	Admin	Member
View group	Yes	Yes	Yes
Invite	Yes	Yes	Optional policy
Edit group	Yes	Yes	No
Manage members	Yes	Yes	No
Change owner/admin	Yes	Limited	No
Delete group	Yes	No	No
Leave group	Yes*	Yes	Yes

* Owner cannot leave without transferring ownership or deleting the group.

CRUD flows

- Create group
- View group
- Edit group
- Archive/delete group
- Add/remove member
- Change member role
- Leave group

UI details

- Member avatars with initials fallback
- Role chip
- Member count
- Danger-zone section for destructive actions
- Confirmation dialogs for removal/delete
- Empty member state should be impossible for a valid group but handle gracefully

Acceptance criteria

- Owner can create a group.
- Members see only groups they belong to.
- Admin cannot perform owner-only actions.
- Deleting/archiving group is protected by rules.
- Member removal updates realtime member list.